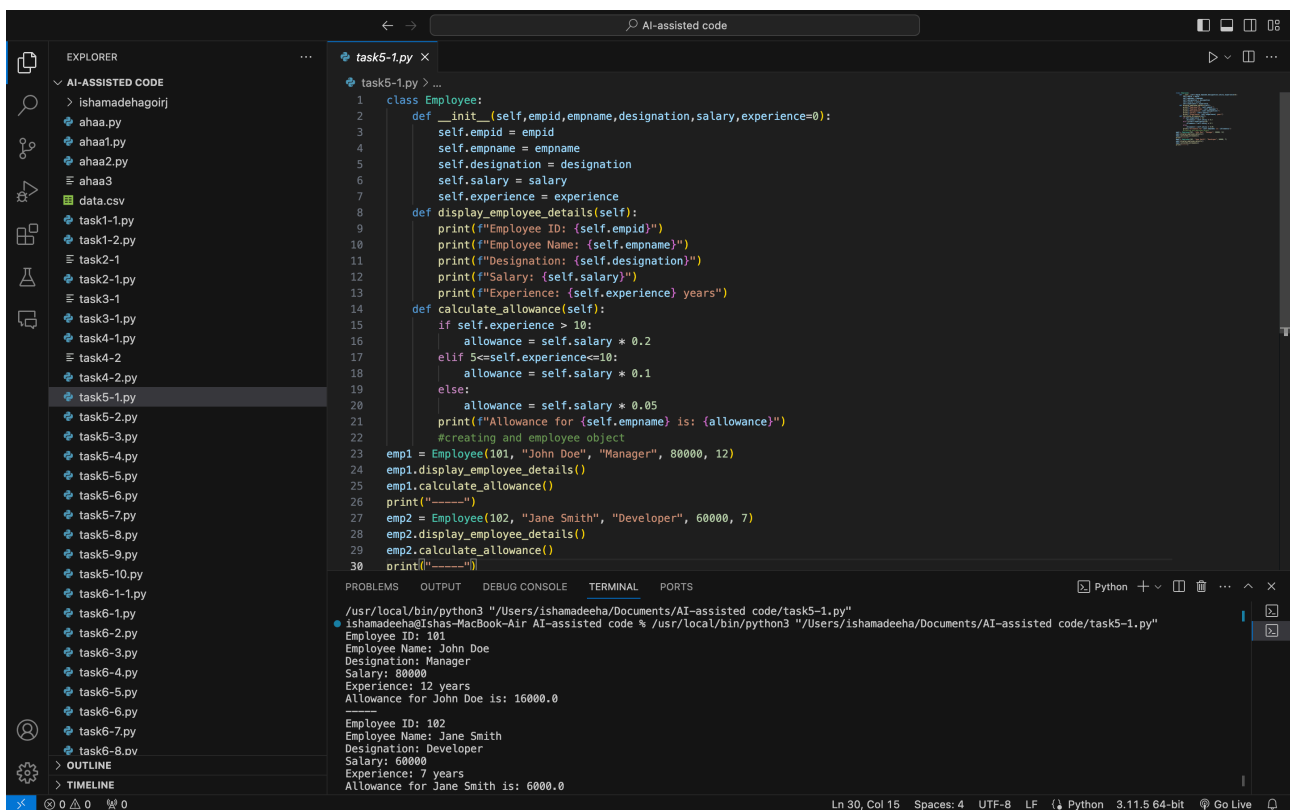


ASSIGNMENT- 5.1 & 6.1

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BATCH-01

TASK-1:



The screenshot displays a Visual Studio Code editor with a Python file named `task5-1.py` open. The file contains a class `Employee` with methods for initialization, displaying details, and calculating allowance based on experience. The terminal output shows the execution of the script, creating two employee objects and displaying their details and calculated allowances.

```
1 class Employee:
2     def __init__(self, empid, empname, designation, salary, experience=0):
3         self.empid = empid
4         self.empname = empname
5         self.designation = designation
6         self.salary = salary
7         self.experience = experience
8     def display_employee_details(self):
9         print(f"Employee ID: {self.empid}")
10        print(f"Employee Name: {self.empname}")
11        print(f"Designation: {self.designation}")
12        print(f"Salary: {self.salary}")
13        print(f"Experience: {self.experience} years")
14    def calculate_allowance(self):
15        if self.experience > 10:
16            allowance = self.salary * 0.2
17        elif 5 <= self.experience <= 10:
18            allowance = self.salary * 0.1
19        else:
20            allowance = self.salary * 0.05
21        print(f"Allowance for {self.empname} is: {allowance}")
22    #creating and employee object
23    emp1 = Employee(101, "John Doe", "Manager", 80000, 12)
24    emp1.display_employee_details()
25    emp1.calculate_allowance()
26    print("-----")
27    emp2 = Employee(102, "Jane Smith", "Developer", 60000, 7)
28    emp2.display_employee_details()
29    emp2.calculate_allowance()
30    print("-----")
```

Terminal Output:

```
/usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-1.py"
Employee ID: 101
Employee Name: John Doe
Designation: Manager
Salary: 80000
Experience: 12 years
Allowance for John Doe is: 16000.0

Employee ID: 102
Employee Name: Jane Smith
Designation: Developer
Salary: 60000
Experience: 7 years
Allowance for Jane Smith is: 6000.0
```

TASK-2:

The screenshot shows the VS Code editor with the file `task5-2.py` open. The file contains a Python class `ElectricityBill` with methods `__init__`, `calculate_bill`, and `display_bill`. The `display_bill` method calls `calculate_bill` and prints customer details and the total amount payable. The terminal output shows the execution of the script, creating two `ElectricityBill` objects and displaying their details.

```
task5-2.py > ElectricityBill > display_bill
1 class ElectricityBill:
2     def __init__(self, customer_id, customer_name, units_consumed):
3         self.customer_id = customer_id
4         self.customer_name = customer_name
5         self.units_consumed = units_consumed
6
7     def calculate_bill(self):
8         if self.units_consumed <= 100:
9             rate = 5
10        elif self.units_consumed <= 300:
11            rate = 10
12        else:
13            rate = 7
14        return self.units_consumed * rate
15
16    def display_bill(self):
17        total_amount = self.calculate_bill()
18        print(f"Customer ID: {self.customer_id}")
19        print(f"Customer Name: {self.customer_name}")
20        print(f"Units Consumed: {self.units_consumed}")
21        print(f"Total Amount Payable: {total_amount:.2f}")
22
23 # Creating an ElectricityBill object
24 bill1 = ElectricityBill(201, "Alice Johnson", 250)
25 bill1.display_bill()
26 print("-----")
27 bill2 = ElectricityBill(202, "Bob Williams", 350)
28 bill2.display_bill()
29 print(f"-----")
```

Terminal Output:

```
Experience: 7 years
Allowance for Jane Smith is: 6000.0
● ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-2.py"
Customer ID: 201
Customer Name: Alice Johnson
Units Consumed: 250
Total Amount Payable: 2500.00
-----
Customer ID: 202
Customer Name: Bob Williams
Units Consumed: 350
Total Amount Payable: 2450.00
-----
```

TASK-3:

The screenshot shows the VS Code editor with the file `task5-3.py` open. The file contains a Python class `Product` with methods `__init__`, `display_product_details`, and `calculate_discount`. The `calculate_discount` method applies different discount rates based on the product category. The terminal output shows the execution of the script, creating two `Product` objects and displaying their details and discounts.

```
task5-3.py > ...
1 class Product:
2     def __init__(self, product_id, product_name, price, category):
3         self.product_id = product_id
4         self.product_name = product_name
5         self.price = price
6         self.category = category
7
8     def display_product_details(self):
9         print(f"Product ID: {self.product_id}")
10        print(f"Product Name: {self.product_name}")
11        print(f"Price: {self.price}")
12        print(f"Category: {self.category}")
13
14    def calculate_discount(self):
15        if self.category.lower() == "electronics":
16            discount = self.price * 0.10
17        elif self.category.lower() == "clothing":
18            discount = self.price * 0.15
19        elif self.category.lower() == "groceries":
20            discount = self.price * 0.05
21        print(f"Discount for {self.product_name} is: {discount}")
22
23 # Creating Product objects
24 prod1 = Product(301, "Smartphone", 5000, "Electronics")
25 prod1.display_product_details()
26 prod1.calculate_discount()
27 print("-----")
28 prod2 = Product(302, "Jeans", 2000, "Clothing")
29 prod2.display_product_details()
30 prod2.calculate_discount()
31 print(f"-----")
```

Terminal Output:

```
● ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-3.py"
Product ID: 301
Product Name: Smartphone
Price: 5000
Category: Electronics
Discount for Smartphone is: 500.0
-----
Product ID: 302
Product Name: Jeans
Price: 2000
Category: Clothing
Discount for Jeans is: 300.0
-----
```

TASK-4:

The screenshot shows a VS Code editor with a file explorer on the left containing files like `ahaa.py`, `data.csv`, and `task1-1.py` through `task6-8.py`. The main editor displays `task5-4.py` with the following Python code:

```
1 class librarybook:
2     def __init__(self,book_id,title,author,borrower,days_late):
3         self.book_id=book_id
4         self.title=title
5         self.author=author
6         self.borrower=borrower
7         self.days_late=days_late
8     def display_details(self):
9         print("Book ID:",self.book_id)
10        print("Title:",self.title)
11        print("Author:",self.author)
12        print("Borrower:",self.borrower)
13        print("Days Late:",self.days_late)
14    def calculate_late_fee(self):
15        if self.days_late<=5:
16            late_fee=self.days_late*5
17        elif 6<= self.days_late<=10:
18            late_fee=self.days_late*7
19        elif self.days_late>10:
20            late_fee=self.days_late*10
21        else:
22            late_fee=0
23        print("Late Fee:",late_fee)
24    book1=librarybook(101,"The Great Gatsby","F. Scott Fitzgerald","Alice",12)
```

The terminal at the bottom shows the execution output:

```
Book ID: 101
Title: The Great Gatsby
Author: F. Scott Fitzgerald
Borrower: Alice
Days Late: 12
Late Fee: 120
Book ID: 202
Title: 1984
Author: George Orwell
Borrower: Bob
Days Late: 4
Late Fee: 20
Book ID: 303
Title: To Kill a Mockingbird
Author: Harper Lee
Borrower: Charlie
Days Late: 7
Late Fee: 49
```

TASK-5:

The screenshot shows a VS Code editor with a file explorer on the left containing files like `ahaa.py`, `data.csv`, and `task1-1.py` through `task6-8.py`. The main editor displays `task5-5.py` with the following Python code:

```
1 def student_report(student_data):
2     report = []
3     for student_name,marks in student_data.items():
4         average_score = sum(marks) / len(marks) if marks else 0
5         status = "Pass" if average_score >= 40 else "Fail"
6         report.append({
7             "name": student_name,
8             "average": round(average_score, 2),
9             "status": status,
10            "marks": marks
11        })
12    return report
13 # Example usage
14 if __name__ == "__main__":
15     students = {
16         "Alice": [85, 90, 78],
17         "Bob": [45, 50, 42],
18         "Charlie": [30, 35, 28],
19         "Diana": [92, 88, 95]
20     }
21     ult = student_report(students)
22     for report in result:
23         print(f'({report["name"]}: Avg={report["average"]}, Status={report["status"]})')
```

The terminal at the bottom shows the execution output:

```
Book ID: 202
Title: 1984
Author: George Orwell
Borrower: Bob
Days Late: 4
Late Fee: 20
Book ID: 303
Title: To Kill a Mockingbird
Author: Harper Lee
Borrower: Charlie
Days Late: 7
Late Fee: 49
Alice: Avg=84.33, Status=Pass
Bob: Avg=45.67, Status=Pass
Charlie: Avg=31.0, Status=Fail
Diana: Avg=91.67, Status=Pass
```

TASK-6:

The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying a project structure under 'AI-ASSISTED CODE'. The file 'task5-6.py' is selected. The main editor window shows the code for a 'TaxiRide' class. The terminal at the bottom displays the output of running the script, showing details for a ride and the total fare.

```
class TaxiRide:
    def __init__(self, ride_id, driver_name, distance_km, waiting_time_min):
        self.ride_id = ride_id
        self.driver_name = driver_name
        self.distance_km = distance_km
        self.waiting_time_min = waiting_time_min

    def display_details(self):
        print(f"Ride ID: {self.ride_id}")
        print(f"Driver Name: {self.driver_name}")
        print(f"Distance: {self.distance_km} km")
        print(f"Waiting Time: {self.waiting_time_min} minutes")

    def calculate_fare(self):
        distance_fare = 0
        remaining_distance = self.distance_km

        # First 10 km at ₹15 per km
        if remaining_distance <= 10:
            distance_fare = remaining_distance * 15
        else:
            distance_fare = 10 * 15
            remaining_distance -= 10

        # Next 20 km at ₹12 per km
        if remaining_distance <= 20:
            distance_fare += remaining_distance * 12
        else:
            distance_fare += 20 * 12
            remaining_distance -= 20

        # Above 30 km at ₹10 per km
        distance_fare += remaining_distance * 10
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + Python 3.11.5 64-bit Go Live

Ln 49, Col 38 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Terminal Output:

```
Diana: Avg=91.67, Status=Pass
Ride ID: TX001
Driver Name: John Doe
Distance: 35 km
Waiting Time: 10 minutes

Total Fare: ₹460
```

TASK-7:

The screenshot shows the VS Code editor with the Explorer sidebar on the left displaying a project structure under 'AI-ASSISTED CODE'. The file 'task5-7.py' is selected. The main editor window shows the code for a 'statistics_subject' function. The terminal at the bottom displays the output of running the script, showing statistics for a list of scores and details for a ride.

```
def statistics_subject(scores_list):
    highest = max(scores_list)
    lowest = min(scores_list)
    average = sum(scores_list) / len(scores_list)
    passed = sum(1 for score in scores_list if score >= 40)
    failed = sum(1 for score in scores_list if score < 40)

    return {
        'highest': highest,
        'lowest': lowest,
        'average': round(average, 2),
        'passed': passed,
        'failed': failed
    }

# Example usage
if __name__ == "__main__":
    scores = [45, 67, 23, 89, 56, 78, 34, 92, 61, 38] + [50] * 50
    result = statistics_subject(scores)
    print(result)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Python + Python 3.11.5 64-bit Go Live

Ln 22, Col 18 Spaces: 4 UTF-8 LF Python 3.11.5 64-bit Go Live

Terminal Output:

```
Days Late: 7
Late Fee: 49
Alice: Avg=84.33, Status=Pass
Bob: Avg=45.67, Status=Pass
Charlie: Avg=31.0, Status=Fail
Diana: Avg=91.67, Status=Pass
Ride ID: TX001
Driver Name: John Doe
Distance: 35 km
Waiting Time: 10 minutes

Total Fare: ₹460
{'highest': 92, 'lowest': 23, 'average': 51.38, 'passed': 57, 'failed': 3}
```

TASK-8:

The screenshot shows a VS Code editor with a file explorer on the left containing various task files. The main editor displays `task5-8.py` with the following code:

```
1 ...
2 Generate a python code to check prime numbers in naive and optimized approach
3 ...
4 def is_prime_naive(n):
5     if n <= 1:
6         return False
7     for i in range(2, n):
8         if n % i == 0:
9             return False
10    return True
11 def is_prime_optimized(n):
12    if n <= 1:
13        return False
14    if n <= 3:
15        return True
16    if n % 2 == 0 or n % 3 == 0:
17        return False
18    i = 5
19    while i * i <= n:
20        if n % i == 0 or n % (i + 2) == 0:
21            return False
22        i += 6
23    return True
24 # Example usage
25 if __name__ == "__main__":
26     number = 29
27     print(f"Naive approach: Is {number} prime? {is_prime_naive(number)}")
28     print(f"Optimized approach: Is {number} prime? {is_prime_optimized(number)}")
```

The terminal at the bottom shows the execution output:

```
Python + Python 3.11.5 64-bit
Alice: Avg=84.33, Status=Pass
Bob: Avg=85.67, Status=Pass
Charlie: Avg=31.0, Status=Fail
Diana: Avg=91.67, Status=Pass
Ride ID: TX001
Driver Name: John Doe
Distance: 35 km
Waiting Time: 10 minutes

Total Fare: ₹460
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-6.py"
{'highest': 92, 'lowest': 23, 'average': 51.38, 'passed': 57, 'failed': 3}
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-7.py"
Naive approach: Is 29 prime? True
Optimized approach: Is 29 prime? True
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

TASK-9:

The screenshot shows a VS Code editor with a file explorer on the left. The main editor displays `task5-9.py` with the following code:

```
1 #generate python code that develop Fibonacci sequence using recursion
2 #provide well commented explanation in each line
3 def fibonacci_recursive(n):
4     # Base case: if n is 0, return 0
5     if n == 0:
6         return 0
7     # Base case: if n is 1, return 1
8     elif n == 1:
9         return 1
10    else:
11        # Recursive case: return the sum of the two preceding numbers
12        return fibonacci_recursive(n - 1) + fibonacci_recursive(n - 2)
13 # Example usage
14 if __name__ == "__main__":
15     # Define the number of terms in the Fibonacci sequence
16     num_terms = 10
17     print(f"Fibonacci sequence up to {num_terms} terms:")
18     # Generate and print the Fibonacci sequence
19     for i in range(num_terms):
20         print(fibonacci_recursive(i), end=" ")
21
```

The terminal at the bottom shows the execution output:

```
Python + Python 3.11.5 64-bit
Diana: Avg=91.67, Status=Pass
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-6.py"
Ride ID: TX001
Driver Name: John Doe
Distance: 35 km
Waiting Time: 10 minutes

Total Fare: ₹460
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-7.py"
{'highest': 92, 'lowest': 23, 'average': 51.38, 'passed': 57, 'failed': 3}
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-8.py"
Naive approach: Is 29 prime? True
Optimized approach: Is 29 prime? True
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-9.py"
Fibonacci sequence up to 10 terms:
0 1 1 2 3 5 8 13 21 34
```

TASK-10:

The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project named 'AI-ASSISTED CODE' with a subdirectory 'ishamadehagairj' containing several Python files. The code editor shows a Python script named 'task5-10.py' with the following code:

```
1 #generate a python code that reads a file and process the data and explain each line meaningfully
2 def read_and_process_file(filename):
3     """
4     Reads a file and processes each line of data.
5
6     Args:
7         filename (str): Path to the file to read
8     """
9     try:
10         with open(filename, 'r') as file:
11             for line_num, line in enumerate(file, 1):
12                 processed_line = line.strip()
13                 if processed_line: # Skip empty lines
14                     print(f"Line (line_num): {processed_line}")
15                     # Add your data processing logic here
16     except FileNotFoundError:
17         print(f"Error: File '{filename}' not found.")
18     except Exception as e:
19         print(f"Error reading file: {e}")
20
21 if __name__ == "__main__":
22     filename = input("Enter the filename: ")
23     read_and_process_file(filename)
```

The terminal output shows the execution of the script for various tasks:

```
Total Fare: ₹460
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-7.py"
{'highest': 92, 'lowest': 23, 'average': 51.38, 'passed': 57, 'failed': 3}
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-8.py"
Naive approach: Is 29 prime? True
Optimized approach: Is 29 prime? True
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-9.py"
Fibonacci sequence up to 10 terms:
0 1 1 2 3 5 8 13 21 34
ishamadeeha@Ishas-MacBook-Air AI-assisted code % /usr/local/bin/python3 "/Users/ishamadeeha/Documents/AI-assisted code/task5-10.py"
Enter the filename: data.csv
Line 1: My name is Isha
Line 2: My right side i have vyshnavi bantu
Line 3: My left side i have kondapalli sruthakeerthi
Line 4: my left left side i have srikaran
ishamadeeha@Ishas-MacBook-Air AI-assisted code %
```

The status bar at the bottom indicates the current file is 'task5-10.py' at line 23, column 36, using UTF-8 encoding and LF line endings, with Python 3.11.5 64-bit as the interpreter.